Semester III **B. Tech CSE (AI)**

BCSAI 305: CLOUD COMPUTING

3L + 0T + 0P + 3C **MM 100**

Unit 1: Understanding cloud computing

Introduction to Cloud Computing -Benefits and Drawbacks - Types of Cloud Service Development - Deployment models

Unit 2: Cloud Architecture Technology and Architectural Requirements

The Business Case for Clouds -Hardware and Infrastructure – Accessing the cloud – Cloud Storage – Standards- Software as a Service – Discovering Cloud Services Development tools. Three Layered Architectural Requirement - Provider Requirements - Service Centric Issues - Interoperability – QoS.

Unit 3: Fault Tolerance

Fault Tolerance - Data Management Storage and Processing – Virtualization Management - Scalability - Load Balancing - Cloud Deployment for Enterprises - User Requirement - Comparative Analysis of Requirement.

Unit 4: Security Management in Cloud

Security Management Standards - Security Management in the Cloud Availability Management – SaaS Availability Management - PaaS Availability Management – IaaS Availability Management - Access Control - Security Vulnerability, Patch, and Configuration Management – Privacy in Cloud- The Key Privacy Concerns in the Cloud - Security in Cloud Computing

Unit 5: Virtualization

Objectives - Benefits - Virtualization Technologies – Data Storage Virtualization – Storage Virtualization – Improving Availability using Virtualization - Improving Performance using Virtualization Improving Capacity using Virtualization.

Text /Reference Books:

1. David S Linthicum, “Cloud Computing and SOA Convergence in your Enterprise A Step by Step Guide”, Addison Wesley Information Technology Series.
2. Anthony T Velte, Toby J.Velte, Robert Elsenpeter, “Cloud computing A Practical Approach “, Tata McGraw Hill Publication
3. Tim Mather, Subra Kumaraswamy, Shahed Latif, “Cloud Security and Privacy
4. An Enterprise Perspective on Risks and Compliance” , O’Reilly Publications, First Edition
5. Michael Miller, “Cloud Computing – Web-Based Applications that Change the Way You Work and Collaborate Online”, Pearson Education, New Delhi, 2009.
6. Cloud Computing Specialist Certification Kit – Virtualization Study Guide